ACN

https://chatgpt.com/share/6994822b-ffc8-8008-adf1-d9e9eff5f9dc

🔷 Q1. (Chapter-1: Introduction)

Explain the TCP/IP layered architecture with a neat diagram. Also explain the function of each layer.

🔷 STEP-1: THEORY EXPLANATION (FROM ZERO LEVEL)

✅ Intuition (Simple Idea)

When we send a message on WhatsApp or open a website, the data does not go directly.

It passes through different stages (layers).
Each layer has a specific job.

Like sending a courier:

- You write letter → Application
- Put in envelope → Transport
- Address written → Network
- Courier delivery → Link/Physical

✅ Basic Explanation

TCP/IP model divides networking into 4 layers:

1. Application Layer
2. Transport Layer
3. Network Layer
4. Link Layer

Each layer performs a specific function.

✅ Exam-Level Definition

TCP/IP architecture is a layered protocol stack used for communication over the Internet, where each layer provides specific services to the upper layer.

✅ Real-Life Analogy

Sending a parcel:

Real Life	Networking
Write message	Application Layer
Pack securely	Transport Layer
Address written	Network Layer
Delivery truck	Link Layer

✅ Practical Use

Every website, email, video call works using TCP/IP model.

✅ Important Exam Writing Points

- Write 4 layers in order.
- Draw layered diagram.
- Mention key protocols in each layer.

🔹 STEP-2: DIAGRAM

```
+-----------------------+
|  Application Layer    |
| (HTTP, FTP, DNS)      |
+-----------------------+
|   Transport Layer     |
| (TCP / UDP)           |
+-----------------------+
|    Network Layer      |
| (IP, Routing)         |
+-----------------------+
|     Link Layer        |
| (Ethernet, WiFi)      |
+-----------------------+
```

🔹 Explanation of Labels

- Application → User-level programs
- Transport → Process-to-process communication
- Network → Host-to-host delivery
- Link → Physical transmission

🔹 STEP-3: FORMULAS

Delay formula (important in TCP/IP):

Total Delay =
Processing + Queueing + Transmission + Propagation

Where:

- Processing Delay = Time to check errors
- Queueing Delay = Waiting in router
- Transmission Delay = Packet size / Bandwidth
- Propagation Delay = Distance / Speed

🔹 STEP-4: NUMERICAL

If packet size = 1000 bits
Bandwidth = 1 Mbps

Transmission Delay = L / R

= 1000 / 1,000,000
= 0.001 sec
= 1 ms

🔹 STEP-5: FINAL ANSWER

👉 TCP/IP model has 4 layers. Each layer performs a specific networking function from application to physical delivery.

🔹 STEP-6: EXAM TIP

- Always draw diagram.
- Write protocols example.
- Mention service provided.
- Avoid confusing with OSI 7-layer model.

🔹 Q2. (Chapter-2: Application Layer)

Explain Client–Server architecture and Peer-to-Peer architecture with diagram. Compare both.

🔹 STEP-1: THEORY

✅ Intuition

Two ways computers communicate:

1. One big server serves everyone (Client-Server)
2. Everyone shares with everyone (P2P)

✅ Basic Explanation

Client-Server:

- One central server
- Clients request data

Peer-to-Peer:

- No central server
- All nodes equal

✅ Exam Definition

Client-Server is a centralized model where clients request services from a dedicated server.

Peer-to-Peer is decentralized architecture where peers directly communicate.

✅ Real-Life Example

Client-Server → Bank server
P2P → BitTorrent file sharing

🔹 STEP-2: DIAGRAM

Client–Server

```
        Client1
           |
Client2 --- Server --- Client3
           |
        Client4
```

Server = Always ON
Clients = Request services

Peer-to-Peer

```
    Peer1 ------ Peer2
      |            |
    Peer3 ------ Peer4
```

All equal. No central server.

🔹 STEP-3: FORMULAS

Server load formula:

Total Load = N × Request Rate

In P2P:

Capacity increases with peers.

🔹 STEP-4: NUMERICAL

If 100 clients request 1 MB file:

Client-server:
Server uploads = 100 MB

P2P:
If each peer shares,
Load distributed → faster.

🔹 STEP-5: FINAL ANSWER

👉 Client-Server is centralized.
👉 P2P is decentralized and scalable.

🔹 STEP-6: EXAM TIP

- Draw both diagrams.
- Write 3 differences.
- Mention scalability.

🔹 Q3. (Chapter-2: Application Layer)

Explain HTTP request-response cycle with diagram. Why is HTTP called stateless?

◆ STEP-1: THEORY

Intuition

Browser asks → Server replies.

Basic Explanation

1. Client sends HTTP Request.
2. Server sends HTTP Response.
3. Connection closed.

Exam Definition

HTTP is an application-layer protocol using TCP for transferring web objects.

Real-Life Analogy

Ordering food:

You order → Waiter serves → Done.

No memory of previous order.

◆ STEP-2: DIAGRAM

```
Client (Browser)                Server
      |  HTTP Request  --------> |
      |                           |
      | <------- HTTP Response   |
      |                           |
Connection Closed
```

◆ STEP-3: FORMULA

HTTP Response Time (Non-persistent)

= 2 RTT + File Transmission Time

Where:

RTT = Round Trip Time

◆ STEP-4: NUMERICAL

If RTT = 100 ms
File time = 50 ms

Total = 2×100 + 50
= 250 ms

🔹 STEP-5: FINAL ANSWER

👉 HTTP is stateless because server does not remember previous requests.

🔹 STEP-6: EXAM TIP

- Write 200 OK example.
- Draw flow diagram.
- Mention TCP port 80.

🔹 Q4. (Chapter-3: Transport Layer)

Explain TCP 3-Way Handshake with diagram.

🔹 STEP-1: THEORY

Intuition

Before communication, both sides agree.

Basic Explanation

3 steps:

1. SYN
2. SYN-ACK
3. ACK

Exam Definition

TCP 3-way handshake is connection establishment process.

Real-Life Example

A: Hello?
B: Yes, I hear you.
A: Good, let's talk.

🔹 STEP-2: DIAGRAM

```
Client                          Server
  | ---- SYN (Seq=x) --------> |
  |                              |
  | <--- SYN+ACK (Seq=y, Ack=x+1)
  |                              |
  | ---- ACK (Ack=y+1) -------> |
Connection Established
```

Label Meaning

SYN = Synchronize
ACK = Acknowledgment
Seq = Sequence number

🔹 STEP-3: FORMULAS

Acknowledgment Number = Sequence + 1

🔹 STEP-4: NUMERICAL

If Seq = 1000

Server sends Ack = 1001

🔹 STEP-5: FINAL ANSWER

👉 TCP uses 3 messages to establish reliable connection.

🔹 STEP-6: EXAM TIP

- Draw handshake diagram.
- Write sequence numbers.
- Mention reliability.

🔹 Q5. (Chapter-3: Transport Layer)

Explain Reliable Data Transfer (rdt3.0) with diagram.

🔹 STEP-1: THEORY

Intuition

If packet lost, resend.

Basic Explanation

rdt3.0 handles:

- Bit errors
- Packet loss
- Timeout retransmission

Exam Definition

rdt3.0 is stop-and-wait protocol using sequence numbers and timer.

Real-Life Example

Sending exam answer sheet:
If teacher doesn't confirm → resend.

🔹 STEP-2: DIAGRAM

```
Sender                                Receiver
  | ---- pkt0 ------------------> |
  |                                 |
  | <---- ACK0 ------------------- |
  |
(If no ACK → Timeout → Retransmit)
```

🔹 STEP-3: FORMULA

Utilization (Stop & Wait):

U = (L/R) / (RTT + L/R)

Where:

L = Packet size
R = Bandwidth

🔹 STEP-4: NUMERICAL

If:

L = 1000 bits
R = 1 Mbps
RTT = 100 ms

L/R = 1000 / 1,000,000 = 0.001 sec

U = 0.001 / (0.1 + 0.001)
= 0.001 / 0.101
≈ 0.0099

= 0.99%

Very inefficient!

🔹 STEP-5: FINAL ANSWER

👉 rdt3.0 ensures reliability using ACK + Timer + Retransmission.

✅ PART 2 — QUESTIONS 6 TO 10

🔹 Q6. (Chapter-3: Transport Layer)

Explain TCP Congestion Control with neat diagram. Describe Slow Start and Congestion Avoidance phases.

🔹 STEP-1: THEORY EXPLANATION (FROM ZERO LEVEL)

✅ **Intuition (Simple Idea)**

If too many cars enter a road → traffic jam.

Similarly:
If too many packets enter network → congestion.

So TCP controls how fast it sends data.

✅ Basic Explanation

TCP uses:

- CWND (Congestion Window)
- SSTHRESH (Slow Start Threshold)

Two main phases:

1. Slow Start
2. Congestion Avoidance

✅ Exam-Level Definition

TCP congestion control is a mechanism to control sender's transmission rate based on network congestion using dynamic window adjustment.

✅ Real-Life Analogy

Water pipe:

Start slowly → increase pressure →
If leakage detected → reduce pressure.

✅ Practical Use

Prevents:

- Packet loss
- Router overflow
- Network collapse

✅ Important Exam Points

- Define CWND
- Define SSTHRESH
- Draw graph
- Explain exponential and linear growth

◆ STEP-2: DIAGRAM

TCP Congestion Window Growth

```
CWND
 ^
 |              /---------
 |             /
 |            /
 |           /
 |          /
 |         /
 |        /
 |       /
 |      /
 |     /
 |    /
 |   /
 |  /
 | /
 |/
 +-----------------------------> Time
     Slow Start     Congestion
                     Avoidance
```

Explanation

- Slow Start → Exponential growth
- After reaching SSTHRESH → Linear growth

◆ STEP-3: FORMULAS WITH SYMBOLS

During Slow Start:

CWND = CWND × 2 (per RTT)

During Congestion Avoidance:

CWND = CWND + 1 MSS per RTT

Where:

CWND = Congestion Window
MSS = Maximum Segment Size

RTT = Round Trip Time

◆ STEP-4: NUMERICAL (Step-by-Step)

Initial:

CWND = 1 MSS
SSTHRESH = 8 MSS

RTT 1 → 1
RTT 2 → 2
RTT 3 → 4
RTT 4 → 8 (Reached threshold)

Now linear:

RTT 5 → 9
RTT 6 → 10

If packet loss at 10:

SSTHRESH = 10/2 = 5
CWND = 1

Restart slow start.

◆ STEP-5: FINAL ANSWER

👉 TCP increases window exponentially then linearly to avoid congestion.

◆ STEP-6: EXAM TIP

- Always draw CWND vs Time graph.
- Mention exponential and linear.
- Write about packet loss behavior.

◆ Q7. (Chapter-3: Transport Layer)

Explain Multiplexing and Demultiplexing with diagram.

◆ STEP-1: THEORY

Intuition

Many apps use internet at same time.

How does computer know which data belongs to which app?

Using Port Numbers.

Basic Explanation

Multiplexing → Sender combines data from multiple apps.
Demultiplexing → Receiver separates data to correct app.

Exam Definition

Multiplexing is process of collecting data from multiple sockets and encapsulating into segments.
Demultiplexing is directing segments to correct socket using port numbers.

Real-Life Example

Post office:

Many letters → one delivery truck → delivered to correct house.

🔹 STEP-2: DIAGRAM

```
Sender Side:

App1 (Port 5000)
App2 (Port 6000)
App3 (Port 7000)
        |
        v
     Transport Layer
        |
        v
       IP

Receiver Side:

        IP
        |
        v
  Transport Layer
```

```
  |      |       |
P5000 P6000 P7000
```

Label Explanation

Port Number = Identifies process
Socket = Combination of IP + Port

🔹 STEP-3: FORMULAS

TCP Socket = (Source IP, Source Port, Dest IP, Dest Port)

UDP Demultiplexing uses:

Destination Port only.

🔹 STEP-4: NUMERICAL

If packet:

Source Port = 5000
Destination Port = 80

Receiver sends to:

Web server (Port 80)

🔹 STEP-5: FINAL ANSWER

👉 Multiplexing combines data from multiple processes.
👉 Demultiplexing delivers to correct process using port numbers.

🔹 STEP-6: EXAM TIP

- Draw both sender & receiver.
- Mention 4-tuple in TCP.
- Write difference between TCP & UDP demux.

🔹 Q8. (Chapter-4: Network Layer – Data Plane)

Explain Longest Prefix Matching with example and diagram.

◆ STEP-1: THEORY

Intuition

Router checks destination IP.

If multiple matches exist → choose most specific match.

Basic Explanation

Forwarding table contains prefixes.

Router matches:

Longest matching prefix bits.

Exam Definition

Longest prefix matching selects forwarding entry with maximum number of matching leading bits.

Real-Life Example

Finding address:

Country → State → City → Street
Most specific wins.

◆ STEP-2: DIAGRAM

Forwarding Table:

```
Prefix                 Interface
1100                   0
110010                 1
11001010               2
Otherwise              3
```

Packet Destination:

110010101010

Matching:

1100 → match
110010 → match
11001010 → match

Longest = 11001010

Forward to Interface 2

🔹 STEP-3: FORMULA

No formula, but:

Match Count = Number of leading identical bits.

🔹 STEP-4: NUMERICAL

Destination = 1100101100

Matches:

1100 → yes
110010 → yes
11001010 → no

Longest = 110010

Interface = 1

🔹 STEP-5: FINAL ANSWER

👉 Router forwards packet using most specific matching prefix.

🔹 STEP-6: EXAM TIP

- Draw table.
- Show matching process.
- Highlight longest prefix.

🔹 Q9. (Chapter-4: Network Layer – Data Plane)

Explain Router Architecture with neat diagram.

🔹 STEP-1: THEORY

Intuition

Router is like traffic police.

It receives packet → checks address → forwards.

Basic Explanation

Router has:

- Input Ports
- Switching Fabric
- Output Ports
- Routing Processor

Exam Definition

Router architecture consists of input ports, switching fabric, output ports and control plane for packet forwarding.

Real-Life Example

Airport:

Entry gate → conveyor belt → exit gate.

◆ STEP-2: DIAGRAM

```
          +-----------------------+
          |  Routing Processor    |
          +-----------------------+


Input Port 1   Input Port 2
     |              |
     v              v
  +-----------------------+
  |   Switching Fabric    |
  +-----------------------+
     ^              ^
     |              |
Output Port 1   Output Port 2
```

Label Explanation

Input Port → Receive packets
Switching Fabric → Moves packets
Output Port → Sends packets
Routing Processor → Runs routing algorithm

🔹 STEP-3: FORMULA

Switching Rate = N × Line Rate

Where:

N = Number of ports

🔹 STEP-4: NUMERICAL

If:

Line rate = 1 Gbps
Ports = 4

Switching capacity required = 4 Gbps

🔹 STEP-5: FINAL ANSWER

👉 Router forwards packets using input ports, switching fabric and output ports.

🔹 STEP-6: EXAM TIP

- Draw full architecture.
- Mention data plane vs control plane.
- Write switching fabric types.

🔹 Q10. (Chapter-2: Application Layer)

Explain DNS Hierarchy with neat diagram.

🔹 STEP-1: THEORY

Intuition

DNS converts:

www.google.com → IP address

Like phone directory.

Basic Explanation

DNS servers are hierarchical:

1. Root DNS
2. TLD DNS (.com)
3. Authoritative DNS

Exam Definition

DNS is distributed hierarchical naming system for mapping domain names to IP addresses.

Real-Life Example

Searching address:

Country → State → City → Street

◆ STEP-2: DIAGRAM

```
           Root DNS
               |
         ---------------
         |             |
       .com          .org
         |
   Authoritative DNS
         |
      www.site.com
```

Label Explanation

Root → Top level
TLD → Domain extension
Authoritative → Stores actual IP

◆ STEP-3: FORMULA

DNS Query Delay:

Total DNS Time = RTT1 + RTT2 + RTT3

Where:

RTT = Round Trip Time

🔹 STEP-4: NUMERICAL

If:

RTT to root = 50ms
RTT to TLD = 40ms
RTT to authoritative = 60ms

Total = 50 + 40 + 60
= 150 ms

🔹 STEP-5: FINAL ANSWER

👉 DNS resolves domain names using hierarchical distributed system.

✅ PART 3 — QUESTIONS 11 TO 15

🔹 Q11. (Chapter-1: Introduction)

Explain the four types of packet delay in a network with diagram and numerical example.

🔹 STEP-1: THEORY EXPLANATION (FROM ZERO LEVEL)

✅ Intuition (Simple Idea)

When you send a message, it does not arrive instantly.

It takes time because:

- Router checks packet
- Packet waits in queue
- Packet travels on wire
- Packet physically moves across distance

These are called delays.

✅ Basic Explanation

There are 4 types of delay:

1. Processing Delay
2. Queueing Delay
3. Transmission Delay
4. Propagation Delay

✅ Exam-Level Definition

Total nodal delay is the sum of processing, queueing, transmission, and propagation delays experienced by a packet at a router.

✅ Real-Life Analogy

Sending parcel:

- Checking parcel → Processing
- Waiting in line → Queueing
- Loading into truck → Transmission
- Driving to city → Propagation

✅ Practical Use

Important for:

- Video streaming
- Gaming
- Voice calls

✅ Important Exam Points

- Write formula
- Explain each delay clearly
- Draw router diagram

◆ STEP-2: DIAGRAM

```
Packet ----> [ Router ] ----> Next Link

Inside Router:

[Processing]
      ↓
[Queue Waiting]
      ↓
[Transmission]
      ↓
[Propagation on Link]
```

Explanation of Labels

Processing → Check header, errors
Queue → Waiting time
Transmission → Push bits into link
Propagation → Travel across medium

◆ STEP-3: FORMULAS WITH SYMBOL NAMES

Total Delay (D) =

D = d_proc + d_queue + d_trans + d_prop

Where:

d_proc = Processing delay
d_queue = Queueing delay
d_trans = Transmission delay
d_prop = Propagation delay

Transmission delay formula:

d_trans = L / R

Where:

L = Packet size (bits)
R = Bandwidth (bits/sec)

Propagation delay:

d_prop = Distance / Speed

◆ STEP-4: NUMERICAL (Step-by-Step)

Given:

Packet size (L) = 1000 bits
Bandwidth (R) = 1 Mbps = 1,000,000 bps
Distance = 100 km = 100,000 m
Speed = 2 × 10^8 m/s

Processing delay = 1 ms
Queue delay = 2 ms

Step 1: Transmission delay

d_trans = L / R
= 1000 / 1,000,000
= 0.001 sec
= 1 ms

Step 2: Propagation delay

d_prop = Distance / Speed
= 100,000 / (2 × 10^8)
= 0.0005 sec
= 0.5 ms

Step 3: Total delay

Total = 1 + 2 + 1 + 0.5
= 4.5 ms

◆ STEP-5: FINAL ANSWER

👉 Total delay is sum of processing, queueing, transmission and propagation delays.

◆ STEP-6: EXAM TIP

- Always write formula first.
- Show unit conversion.
- Don’t confuse transmission with propagation.

◆ Q12. (Chapter-2: Application Layer)

Explain Web Caching with diagram and discuss its advantages.

◆ STEP-1: THEORY

Intuition

Instead of going to main server every time,
use local copy.

Basic Explanation

Web cache (proxy server):

- Stores frequently requested objects
- Reduces Internet traffic
- Improves response time

Exam Definition

Web caching is storing web objects locally to satisfy client requests without contacting origin server.

Real-Life Example

School library keeps copy of textbook instead of ordering from publisher every time.

Practical Use

Used by:

- ISPs
- Companies
- CDNs

Important Exam Points

- Draw client-cache-server diagram
- Mention hit & miss

◆ STEP-2: DIAGRAM

```
Client
   |
```

```
   v
Web Cache
   |
   v
Origin Server
```

Flow:

1. Client asks cache.
2. If found → return.
3. If not → fetch from server.

🔹 STEP-3: FORMULAS

Access link utilization:

Utilization = Data Rate / Link Capacity

Cache Hit Ratio:

Hit Ratio = (Cache Hits / Total Requests)

🔹 STEP-4: NUMERICAL

Given:

Link capacity = 1.5 Mbps
Request rate = 15/sec
Object size = 100 Kbits

Data rate = 15 × 100K
= 1500 Kbits/sec
= 1.5 Mbps

Utilization = 1.5 / 1.5
= 1 (Fully utilized → congestion)

If cache hit ratio = 0.4

New traffic = 60%
= 0.6 × 1.5
= 0.9 Mbps

Now utilization = 0.9 / 1.5
= 0.6

Reduced congestion.

◆ STEP-5: FINAL ANSWER

👉 Web caching reduces delay and bandwidth usage by serving local copies.

◆ STEP-6: EXAM TIP

- Draw diagram clearly.
- Write benefits: reduced delay, reduced traffic.
- Mention hit ratio.

◆ Q13. (Chapter-3: Transport Layer)

Explain UDP protocol with header format diagram.

◆ STEP-1: THEORY

Intuition

UDP is simple and fast.

No connection, no guarantee.

Basic Explanation

Features:

- Connectionless
- Unreliable
- No congestion control
- Low overhead

Exam Definition

UDP is a connectionless transport protocol providing best-effort delivery without reliability guarantees.

Real-Life Example

Sending normal post without tracking.

Practical Use

Used in:

- Video streaming
- DNS
- Online games

◆ STEP-2: DIAGRAM

UDP Header Format:

```
 0      15 16     31
+--------+--------+
| Source | Dest   |
| Port   | Port   |
+--------+--------+
| Length |Checksum|
+--------+--------+
|       Data      |
+-----------------+
```

Label Explanation

Source Port → Sender process
Destination Port → Receiver process
Length → Total UDP segment size
Checksum → Error detection

◆ STEP-3: FORMULAS

Checksum = One's complement sum of 16-bit words.

Length = Header + Data

◆ STEP-4: NUMERICAL

If:

Data = 100 bytes
Header = 8 bytes

Length = 100 + 8
= 108 bytes

🔹 STEP-5: FINAL ANSWER

👉 UDP provides fast but unreliable transport with 8-byte header.

🔹 STEP-6: EXAM TIP

- Draw header format.
- Write port numbers.
- Mention no congestion control.

🔹 Q14. (Chapter-3: Transport Layer)

Derive and explain Stop-and-Wait protocol efficiency.

🔹 STEP-1: THEORY

Intuition

Send one packet → wait for ACK → send next.

Slow but simple.

Basic Explanation

Stop-and-Wait:

- Send 1 packet
- Wait for ACK
- Uses timer

Exam Definition

Stop-and-Wait is a reliable data transfer protocol where sender waits for ACK before sending next packet.

Real-Life Example

Teacher gives one question → waits for answer → gives next.

🔹 STEP-2: DIAGRAM

```
Sender                Receiver
  | ---- pkt1 -----> |
```

```
| <---- ACK1 ----- |
| ---- pkt2 -----> |
| <---- ACK2 ----- |
```

🔹 STEP-3: FORMULA

Efficiency (Utilization):

U = (L/R) / (RTT + L/R)

Where:

L = Packet size
R = Bandwidth
RTT = Round Trip Time

🔹 STEP-4: NUMERICAL

Given:

L = 1000 bits
R = 1 Mbps
RTT = 100 ms

Step 1: L/R

= 1000 / 1,000,000
= 0.001 sec

Step 2:

U = 0.001 / (0.1 + 0.001)
= 0.001 / 0.101
≈ 0.0099

= 0.99%

Very low efficiency.

🔹 STEP-5: FINAL ANSWER

👉 Stop-and-Wait efficiency is very low for large RTT.

🔹 STEP-6: EXAM TIP

- Write formula clearly.
- Explain why inefficient.
- Mention sliding window as improvement.

◆ Q15. (Chapter-4: Network Layer – Data Plane)

Explain Switching Fabric types with diagram.

◆ STEP-1: THEORY

Intuition

Switching fabric moves packet from input to output.

Basic Explanation

Three types:

1. Switching via Memory
2. Switching via Bus
3. Switching via Interconnection Network

Exam Definition

Switching fabric is internal mechanism that transfers packets from input ports to output ports in router.

Real-Life Example

Airport luggage system.

◆ STEP-2: DIAGRAM

(1) Memory

```
Input → CPU Memory → Output
```

Packet copied twice.

---

(2) Bus

```
Input Ports → Shared Bus → Output Ports
```

Single shared medium.

---

(3) Crossbar

```
Inputs
 | \   / |
 |  \ /  |
 |  / \   |
 | /   \  |
Outputs
```

Parallel connections.

🔹 STEP-3: FORMULA

Switching rate must be:

Switching Rate ≥ N × Line Rate

🔹 STEP-4: NUMERICAL

If:

Ports = 4
Line rate = 1 Gbps

Switching capacity needed = 4 Gbps

🔹 STEP-5: FINAL ANSWER

👉 Switching fabric transfers packets inside router using memory, bus or crossbar techniques.

🔹 STEP-6: EXAM TIP

- Draw all three.
- Mention advantages.
- Write about scalability.

🔷 Q16. (Chapter-4: Network Layer – Data Plane)

Explain IP Datagram Format with neat diagram and explain each field.

🔷 STEP-1: THEORY EXPLANATION (FROM ZERO LEVEL)

✅ Intuition (Simple Idea)

When data travels in the Internet, it is wrapped inside an IP packet.

This packet contains:

- Address
- Control information
- Data

Like a courier envelope.

✅ Basic Explanation

IP datagram consists of:

- Header (control info)
- Data (payload)

Header contains fields like:

- Version
- Header Length
- Total Length
- Source IP
- Destination IP
- TTL
- Protocol

✅ Exam-Level Definition

IP datagram is a network-layer packet that encapsulates transport-layer segments and contains addressing and control information for routing.

✅ Real-Life Analogy

Courier envelope:

- Sender address → Source IP
- Receiver address → Destination IP
- Expiry limit → TTL
- Parcel content → Data

✅ Practical Use

Routers read IP header to forward packet.

✅ Important Exam Points

- Draw header diagram
- Explain TTL and Protocol
- Mention IPv4 (32-bit address)

🔹 STEP-2: DIAGRAM (IP Datagram Header)

```
 0                   15 16                   31
+--------+--------+--------+----------------+
|Version |  IHL   |  Type  |  Total Length  |
+--------+--------+--------+----------------+
| Identification   |Flags| Fragment Offset  |
+-----------------------------------------+
|   TTL   | Protocol |     Header Checksum |
+-----------------------------------------+
|          Source IP Address (32 bits)     |
+-----------------------------------------+
|       Destination IP Address (32 bits)   |
+-----------------------------------------+
|                Data (Payload)            |
+-----------------------------------------+
```

🔹 Explanation of Fields

Version → IP version (4 for IPv4)
IHL → Internet Header Length
Total Length → Entire packet size

TTL (Time To Live) → Limits packet life
Protocol → TCP (6), UDP (17)
Checksum → Error detection
Source IP → Sender address
Destination IP → Receiver address

🔹 STEP-3: FORMULAS

Total Length = Header + Data

Header Length (IHL) × 4 bytes

🔹 STEP-4: NUMERICAL

If:

IHL = 5

Header size = 5 × 4
= 20 bytes

If Data = 1000 bytes

Total Length = 20 + 1000
= 1020 bytes

🔹 STEP-5: FINAL ANSWER

👉 IP datagram contains header fields used for routing and payload for data transmission.

🔹 STEP-6: EXAM TIP

- Draw full header.
- Explain TTL carefully.
- Mention 32-bit IP address.

🔹 Q17. (Chapter-3: Transport Layer)

Explain Flow Control in TCP with diagram.

🔹 STEP-1: THEORY

✅ Intuition

If receiver is slow, sender must slow down.

Otherwise buffer overflow happens.

✅ Basic Explanation

TCP uses:

Receiver Window (RWND)

Receiver tells sender:

“How much data I can accept.”

✅ Exam Definition

TCP flow control prevents sender from overwhelming receiver by limiting amount of unacknowledged data using advertised window.

✅ Real-Life Example

Teacher says:

“I can check only 5 copies at a time.”

✅ Practical Use

Prevents:

- Receiver buffer overflow
- Data loss

Important Exam Points

- Mention sliding window
- Explain advertised window

🔹 STEP-2: DIAGRAM

```
Sender                               Receiver
  |---- Data (Window size 4) ---->|
  |<---- ACK + RWND=4 -----------|
```

Sliding Window View:

```
|---- Sent ----|---- Allowed ----|---- Not Allowed ----|
        ↑
      ACKed
```

Label Explanation

RWND = Receiver Window
Sender can send only RWND bytes

◆ STEP-3: FORMULA

Effective Window Size = min(CWND, RWND)

Where:

CWND = Congestion Window
RWND = Receiver Window

◆ STEP-4: NUMERICAL

If:

CWND = 8 MSS
RWND = 5 MSS

Effective window = min(8,5)
= 5 MSS

Sender can send only 5 segments.

◆ STEP-5: FINAL ANSWER

👉 TCP flow control ensures sender does not overflow receiver buffer.

◆ STEP-6: EXAM TIP

- Draw sliding window.
- Write min(CWND, RWND).
- Mention receiver advertisement.

◆ Q18. (Chapter-2: Application Layer)

Explain Cookies in HTTP with diagram. How are they used in advertising?

◆ STEP-1: THEORY

Intuition

Website remembers you using small data called cookie.

Basic Explanation

Cookie has 4 components:

1. Cookie in HTTP response
2. Cookie in HTTP request
3. Cookie file in browser
4. Backend database

Exam Definition

Cookie is small piece of data stored at client to maintain state in stateless HTTP protocol.

Real-Life Example

Restaurant remembers your favorite dish.

Practical Use

Used in:

- Login sessions
- Shopping carts
- Targeted ads

Important Exam Points

- Draw 3-site advertising example
- Mention privacy

◆ STEP-2: DIAGRAM

```
Client Browser
     |
     | HTTP Request
     v
Website Server
     |
```

```
     | HTTP Response + Cookie ID
     v
Client stores Cookie

Next visit:
Client sends Cookie ID
Server identifies user
```

Advertising Case:

```
User → News.com → AdServer.com
           |
           → Cookie shared
```

Ad server tracks browsing behavior.

◆ STEP-3: FORMULA

No mathematical formula.

Cookie ID = Unique identifier stored in DB.

◆ STEP-4: NUMERICAL (Conceptual Example)

If:

User visits 5 websites with same Ad server

Ad server records:

UserID123 → Sports, Shopping, Movies

Shows targeted ads.

◆ STEP-5: FINAL ANSWER

👉 Cookies help maintain user state and enable targeted advertising.

◆ STEP-6: EXAM TIP

- Draw flow diagram.
- Mention stateless HTTP.
- Write privacy concern.

◆ Q19. (Chapter-4: Network Layer – Data Plane)

Explain Network Service Model (Best Effort vs QoS).

◆ STEP-1: THEORY

Intuition

Internet mostly gives “best effort”.

No guarantee of speed or delay.

Basic Explanation

Service Models:

1. Best Effort
2. Guaranteed Bandwidth
3. Guaranteed Delay

Internet uses Best Effort.

Exam Definition

Network service model defines guarantees provided to packets during transmission.

Real-Life Example

Normal road (Best effort)
VIP road (Guaranteed)

Practical Use

Real-time apps need QoS.

◆ STEP-2: DIAGRAM

```
Best Effort Model:

Host → Router → Router → Host
(No guarantee)

QoS Model:
```

```
Host → Router (Priority Queue) → Host
(Guaranteed bandwidth)
```

🔹 STEP-3: FORMULA

Throughput = Data / Time

No strict QoS guarantee in best effort.

🔹 STEP-4: NUMERICAL

If link = 10 Mbps

Two users:

User A = 6 Mbps
User B = 4 Mbps

Best effort:

Bandwidth may vary.

QoS:

User A guaranteed 6 Mbps.

🔹 STEP-5: FINAL ANSWER

👉 Internet provides best effort service without strict guarantees.

🔹 STEP-6: EXAM TIP

- Compare best effort & QoS.
- Write advantages of simplicity.
- Mention DiffServ briefly.

🔹 Q20. (Chapter-1: Introduction)

Explain Packet Switching vs Circuit Switching with diagram and comparison.

🔹 STEP-1: THEORY

Intuition

Two ways to send data:

1. Reserve full path (Circuit)
2. Send in packets (Packet)

Basic Explanation

Circuit Switching:

- Dedicated path
- Fixed bandwidth

Packet Switching:

- Data divided into packets
- Shared network

Exam Definition

Packet switching transmits data in small packets through shared network resources, whereas circuit switching reserves dedicated resources.

Real-Life Example

Circuit → Phone call
Packet → Internet browsing

◆ STEP-2: DIAGRAM

Circuit Switching:

```
A ====== B ====== C
(Reserved path)
```

Packet Switching:

```
A → R1 → R2 → C
Packets take different paths
```

◆ STEP-3: FORMULA

Packet transmission delay:

L / R

◆ STEP-4: NUMERICAL

Link = 1 Gbps
Each user needs 100 Mbps

Circuit switching:

Max users = 1Gbps / 100Mbps
= 10 users

Packet switching:

More users possible if active 10% time.

◆ STEP-5: FINAL ANSWER

👉 Packet switching is efficient and scalable; circuit switching provides guaranteed bandwidth.

◆ STEP-6: EXAM TIP

- Draw both diagrams.
- Write 4 comparison points.
- Mention resource sharing.

✅ PART 5 — QUESTIONS 21 TO 25

◆ Q21. (Chapter-3: Transport Layer)

Explain TCP Connection Termination (Four-Way Handshake) with neat diagram.

🔹 STEP-1: THEORY EXPLANATION (FROM ZERO LEVEL)

✅ Intuition (Simple Idea)

When two people finish talking on phone:

Person A: “I’m done.”
Person B: “Okay, I’m done too.”

TCP also closes connection politely.

✅ Basic Explanation

TCP termination uses 4 steps:

1. FIN
2. ACK
3. FIN
4. ACK

Because TCP is full duplex (data flows both sides).

✅ Exam-Level Definition

TCP connection termination is a four-step handshake process using FIN and ACK flags to gracefully close a connection.

✅ Real-Life Analogy

Conversation:

A: I finished talking.
B: Okay.
B: I finished too.
A: Okay.

✅ Practical Use

Ensures:

- All data transmitted
- No data loss

- Resources freed properly

✅ Important Exam Points

- Mention full-duplex
- Draw 4-message exchange
- Explain TIME_WAIT

🔹 STEP-2: DIAGRAM

```
Client                                    Server
  | ---- FIN (Seq=x) --------------> |
  |                                   |
  | <---- ACK (Ack=x+1) ------------- |
  |                                   |
  | <---- FIN (Seq=y) --------------- |
  |                                   |
  | ---- ACK (Ack=y+1) -------------> |
Connection Closed
```

Label Explanation

FIN = Finish
ACK = Acknowledgment
Seq = Sequence Number

🔹 STEP-3: FORMULAS

Acknowledgment Number = Sequence Number + 1

🔹 STEP-4: NUMERICAL

If Client sends:

FIN with Seq = 2000

Server replies:

ACK = 2001

Server sends:

FIN with Seq = 3000

Client replies:

ACK = 3001

🔹 STEP-5: FINAL ANSWER

👉 TCP uses four-way handshake to terminate connection gracefully.

🔹 STEP-6: EXAM TIP

- Draw 4 steps clearly.
- Write FIN and ACK flags.
- Mention full-duplex.

🔹 Q22. (Chapter-4: Network Layer – Data Plane)

Explain NAT (Network Address Translation) with diagram and working.

🔹 STEP-1: THEORY

✅ Intuition

Inside home:

Private IP addresses.

Outside Internet:

Public IP address.

NAT converts private IP to public IP.

✅ Basic Explanation

NAT allows multiple devices in LAN to share one public IP.

Router changes:

Source IP
Source Port

✅ Exam Definition

NAT is a technique where private IP addresses are translated into public IP addresses by modifying packet headers at router.

✅ Real-Life Analogy

Apartment building:

Many people share one building address.

✅ Practical Use

- Saves IP addresses
- Provides basic security

Important Exam Points

- Draw before and after translation
- Mention port mapping

🔹 STEP-2: DIAGRAM

Before NAT:

```
Host (192.168.1.2:5000)
        |
        v
NAT Router (Public IP: 100.10.10.5)
        |
        v
Internet Server (8.8.8.8)
```

After Translation:

```
Private Packet:
Src IP: 192.168.1.2
Src Port: 5000

Translated Packet:
Src IP: 100.10.10.5
Src Port: 40000
```

NAT Table:

```
Inside IP      Inside Port   Public Port
192.168.1.2       5000          40000
```

◆ STEP-3: FORMULA

No direct formula.

Port mapping:

(New Public Port) ↔ (Private IP, Private Port)

◆ STEP-4: NUMERICAL

If 3 devices:

192.168.1.2
192.168.1.3
192.168.1.4

All use same public IP:

100.10.10.5

NAT assigns different ports:

40000
40001
40002

◆ STEP-5: FINAL ANSWER

👉 NAT translates private IP addresses into public IP addresses using port mapping.

◆ STEP-6: EXAM TIP

- Draw NAT table.
- Show IP and port changes.
- Mention IPv4 shortage.

◆ Q23. (Chapter-3: Transport Layer)

Explain Sliding Window Protocol with diagram.

◆ STEP-1: THEORY

✅ **Intuition**

Instead of sending 1 packet and waiting,
send multiple packets before waiting.

✅ Basic Explanation

Sliding Window allows:

- Multiple unacknowledged packets
- Better efficiency

✅ Exam Definition

Sliding window protocol allows sender to transmit multiple packets within a window size before receiving acknowledgment.

✅ Real-Life Analogy

Teacher gives 5 questions at once instead of one by one.

Practical Use

Used in TCP for efficiency.

Important Exam Points

- Draw window diagram
- Explain window sliding

🔹 STEP-2: DIAGRAM

```
Sequence Numbers:
0 1 2 3 4 5 6 7 8 9

Window Size = 4

[0 1 2 3] 4 5 6 7 8 9

After ACK 0:

0 [1 2 3 4] 5 6 7 8 9
```

Label Explanation

Window Size = Number of packets allowed
ACK shifts window forward

◆ STEP-3: FORMULA

Window Size = Bandwidth × RTT

(For full utilization)

◆ STEP-4: NUMERICAL

Given:

Bandwidth = 1 Mbps
RTT = 100 ms = 0.1 sec

Window Size = 1,000,000 × 0.1
= 100,000 bits

If packet size = 1000 bits

Packets in window = 100,000 / 1000
= 100 packets

◆ STEP-5: FINAL ANSWER

👉 Sliding window improves efficiency by allowing multiple outstanding packets.

◆ STEP-6: EXAM TIP

- Draw window movement.
- Write formula B×RTT.
- Mention efficiency improvement.

◆ Q24. (Chapter-2: Application Layer)

Explain Non-Persistent vs Persistent HTTP with diagram and comparison.

◆ STEP-1: THEORY

Intuition

Non-Persistent:

Open connection → send 1 object → close.

Persistent:

Open once → send many objects → close.

Basic Explanation

Non-Persistent:

2 RTT per object.

Persistent:

1 RTT for multiple objects.

Exam Definition

Persistent HTTP allows multiple objects to be sent over single TCP connection, whereas non-persistent HTTP uses separate connection per object.

Real-Life Example

Non-persistent → Reopening door each time
Persistent → Keep door open

🔹 STEP-2: DIAGRAM

Non-Persistent:

```
Client → TCP Open → Object1 → Close
Client → TCP Open → Object2 → Close
```

Persistent:

```
Client → TCP Open
         Object1
         Object2
         Object3
         Close
```

🔹 STEP-3: FORMULA

Non-Persistent Response Time:

= 2RTT + Transmission Time

Persistent:

≈ 1RTT + Transmission Time

🔹 STEP-4: NUMERICAL

If:

RTT = 100 ms
Transmission = 50 ms

Non-persistent (3 objects):

Each = 250 ms

Total = 750 ms

Persistent:

1 RTT + total transmission

= 100 + 150
= 250 ms

🔹 STEP-5: FINAL ANSWER

👉 Persistent HTTP reduces delay by reusing TCP connection.

🔹 STEP-6: EXAM TIP

- Draw both diagrams.
- Write 2RTT vs 1RTT.
- Mention HTTP/1.1 uses persistent.

🔹 Q25. (Chapter-4: Network Layer – Data Plane)

Explain Output Port Queuing and Scheduling Policies with diagram.

🔹 STEP-1: THEORY

Intuition

If packets arrive faster than link speed,
they wait in queue.

Basic Explanation

Output port has:

- Buffer
- Scheduling policy

Policies:

1. FIFO
2. Priority
3. Round Robin

Exam Definition

Output port queuing occurs when arrival rate exceeds output link capacity, requiring buffer and scheduling mechanism.

Real-Life Example

Supermarket billing counter.

Important Exam Points

- Draw queue
- Explain scheduling types

🔹 STEP-2: DIAGRAM

```
Packets Arriving
      ↓
   [ Output Buffer ]
   | P1 |
   | P2 |
   | P3 |
      ↓
   Transmission Line
```

Priority Scheduling:

```
High Queue → Sent First
Low Queue  → Sent Later
```

Round Robin:

```
Q1 → Q2 → Q3 → Q1 → Q2 → Q3
```

◆ STEP-3: FORMULA

Queueing delay depends on:

Traffic Intensity (ρ)

ρ = (Arrival Rate × Packet Size) / Bandwidth

If $\rho \to 1$
Delay increases.

◆ STEP-4: NUMERICAL

If:

Arrival rate = 100 packets/sec
Packet size = 1000 bits
Bandwidth = 100,000 bits/sec

$\rho = (100 \times 1000) / 100{,}000$
$= 100{,}000 / 100{,}000$
$= 1$

Very high delay.

◆ STEP-5: FINAL ANSWER

👉 Output port queuing occurs due to congestion and scheduling policies decide packet transmission order.

◆ STEP-6: EXAM TIP

- Draw queue diagram.
- Mention FIFO, Priority, RR.
- Write traffic intensity formula.

🔹 Q26. (Chapter-3: Transport Layer)

Explain Internet Checksum with working and example calculation.

🔹 STEP-1: THEORY EXPLANATION (FROM ZERO LEVEL)

✅ Intuition (Simple Idea)

When data travels in network, bits may flip (0 becomes 1).

So we need a method to detect errors.

Checksum helps detect errors.

✅ Basic Explanation

- Sender divides data into 16-bit words.
- Adds them using 1’s complement addition.
- Takes 1’s complement of result.
- Sends it as checksum.
- Receiver repeats calculation.
- If result = all 1’s → No error.

✅ Exam-Level Definition

Internet checksum is an error detection technique that computes the 1’s complement sum of 16-bit words in a segment.

✅ Real-Life Analogy

Like adding numbers and verifying total in bank.

✅ Practical Use

Used in:

- UDP
- TCP
- IP

✅ Important Exam Points

- Mention 16-bit addition
- Explain 1’s complement
- Show wrap-around carry

🔹 STEP-2: DIAGRAM

```
Sender Side:

Data → Divide into 16-bit words
      ↓
Add all words
      ↓
Take 1’s complement
      ↓
Attach as Checksum

Receiver Side:

Add all words including checksum
If result = 1111111111111111 → No error
```

🔹 STEP-3: FORMULA

Checksum = 1’s Complement (Sum of 16-bit words)

🔹 STEP-4: NUMERICAL (Step-by-Step)

Given 2 words:

Word1 = 11001010 00001111
Word2 = 10101010 00000011

Step 1: Add

11001010 00001111
+10101010 00000011

1 01110100 00010010

Step 2: Wrap carry

Carry = 1

01110100 00010010
+00000000 00000001

01110100 00010011

Step 3: Take 1’s complement

10001011 11101100

This is checksum.

🔹 STEP-5: FINAL ANSWER

👉 Internet checksum detects bit errors using 1’s complement addition of 16-bit words.

🔹 STEP-6: EXAM TIP

- Always show binary addition.
- Mention wrap-around carry.
- Write 1’s complement clearly.

🔹 Q27. (Chapter-3: Transport Layer)

Explain rdt2.0 and its problem (garbled ACK) with diagram.

🔹 STEP-1: THEORY

✅ Intuition

If channel has bit errors:

Receiver sends ACK or NAK.

But what if ACK itself gets corrupted?

✅ Basic Explanation

rdt2.0:

- Uses checksum
- Uses ACK and NAK
- Retransmits on NAK

Problem:

ACK/NAK may get corrupted.

✅ Exam Definition

rdt2.0 is reliable data transfer protocol for channels with bit errors using ACK/NAK.

Real-Life Analogy

Teacher says “Correct” or “Wrong”.

If student hears noise → confusion.

🔹 STEP-2: DIAGRAM

```
Sender                      Receiver
  | ---- pkt0 --------> |
  |                       |
  | <---- ACK --------- |
```

Problem case:

```
Sender                      Receiver
  | ---- pkt0 --------> |
  |                       |
  | <---- ACK (corrupt) |
Sender doesn't know what happened!
```

🔹 STEP-3: FORMULA

Sequence number added in rdt2.1 to solve problem.

Seq = 0 or 1

🔹 STEP-4: NUMERICAL (Concept)

If packet 0 sent.

ACK corrupted.

Sender retransmits packet 0.

Without sequence number → receiver may deliver duplicate.

🔹 STEP-5: FINAL ANSWER

👉 rdt2.0 fails when ACK/NAK is corrupted because sender cannot detect duplicate situation.

🔹 STEP-6: EXAM TIP

- Draw ACK corruption scenario.
- Mention fatal flaw.
- Explain need for sequence numbers.

🔹 Q28. (Chapter-4: Network Layer – Data Plane)

Explain Input Port Queuing and Head-of-Line (HOL) blocking with diagram.

🔹 STEP-1: THEORY

✅ Intuition

If first packet in queue is blocked, others behind cannot move.

Like traffic jam at signal.

✅ Basic Explanation

Input port queuing occurs when:

Switch fabric slower than input rate.

HOL blocking:

Front packet blocks others even if they can go.

✅ Exam Definition

Head-of-Line blocking occurs when first packet in input queue prevents others from being forwarded despite free output ports.

Real-Life Analogy

Single-lane road:

Truck in front blocks all cars.

◆ STEP-2: DIAGRAM

```
Input Queue:

[P1 → Output 1]
[P2 → Output 2]
[P3 → Output 3]

If Output 1 busy,
P1 waits.
P2 & P3 also wait.
```

◆ STEP-3: FORMULA

Switching Rate < N × Line Rate
→ Queue builds.

◆ STEP-4: NUMERICAL

If:

Input rate = 4 Gbps
Switch capacity = 3 Gbps

Excess = 1 Gbps

Queue increases over time.

◆ STEP-5: FINAL ANSWER

👉 HOL blocking reduces performance because packets behind cannot move even if their path is free.

◆ STEP-6: EXAM TIP

- Draw queue clearly.
- Mention switch speed condition.

- Write definition precisely.

◆ Q29. (Chapter-2: Application Layer)

Explain Transport Layer Service Requirements of Applications.

◆ STEP-1: THEORY

✅ Intuition

Different apps need different network services.

Example:

Email needs reliability.
Video needs low delay.

✅ Basic Explanation

4 Requirements:

1. Data Integrity
2. Timing
3. Throughput
4. Security

✅ Exam Definition

Transport service requirements define reliability, timing, throughput and security needs of application-layer protocols.

Real-Life Analogy

Food delivery:

Some need fast (pizza).
Some need accurate (documents).

◆ STEP-2: DIAGRAM

```
Application
    |
    v
```

```
Transport Layer
 (TCP or UDP)

App Requirement → Choose TCP/UDP
```

🔹 STEP-3: FORMULAS

Throughput = Data / Time

Delay sensitive apps require:

Low RTT.

🔹 STEP-4: NUMERICAL

If:

Video needs 2 Mbps
Available bandwidth = 1 Mbps

App will lag.

If Email 1 MB file:

Needs reliability more than speed.

🔹 STEP-5: FINAL ANSWER

👉 Applications choose TCP or UDP based on reliability, delay, throughput and security needs.

🔹 STEP-6: EXAM TIP

- Write 4 requirements.
- Give examples (Email, Video).
- Mention TCP vs UDP comparison.

🔹 Q30. (Chapter-1: Introduction)

Explain Throughput and Bottleneck Link with numerical example.

🔹 STEP-1: THEORY

✅ Intuition

Throughput = Actual speed of data transfer.

Bottleneck = Slowest link in path.

✅ Basic Explanation

If path has:

10 Mbps → 5 Mbps → 20 Mbps

Maximum throughput = 5 Mbps.

✅ Exam Definition

Throughput is the rate at which data is successfully delivered over network. Bottleneck link determines maximum throughput.

Real-Life Analogy

Water pipe:

Smallest pipe limits flow.

🔹 STEP-2: DIAGRAM

```
Host A --10Mbps-- R1 --5Mbps-- R2 --20Mbps-- Host B

Bottleneck = 5 Mbps
```

🔹 STEP-3: FORMULA

Throughput = Minimum Link Capacity

🔹 STEP-4: NUMERICAL

Given:

Link1 = 10 Mbps
Link2 = 5 Mbps
Link3 = 20 Mbps

Throughput = min(10,5,20)
= 5 Mbps

If sending 50 MB file:

Time = Data / Throughput

= 50 MB / 5 Mbps

Convert:

50 MB = 400 Mb

Time = 400 / 5
= 80 seconds

🔹 STEP-5: FINAL ANSWER

👉 Throughput is limited by bottleneck link which is slowest link in path.

🔹 STEP-6: EXAM TIP

- Draw path diagram.
- Always choose minimum link.
- Convert units properly (MB to Mb).

🔹 Q1. (Chapter-3: Transport Layer)

Explain TCP in detail. Discuss connection establishment, reliable data transfer, flow control, congestion control and connection termination with diagrams.

🔹 STEP-1: THEORY (FROM ZERO LEVEL)

✅ Intuition

TCP is like a very careful delivery service.

It ensures:

- No data loss
- Correct order
- No congestion
- No receiver overflow

✅ What TCP Provides

1. Connection-oriented communication
2. Reliable data transfer
3. Flow control
4. Congestion control
5. Full duplex communication

🔹 1 TCP CONNECTION ESTABLISHMENT

3-Way Handshake

```
Client                         Server
  | ---- SYN (Seq=x) --------> |
  |                             |
  | <--- SYN+ACK (Seq=y) ------ |
  |                             |
  | ---- ACK (Ack=y+1) -------> |
Connection Established
```

Purpose:

- Synchronize sequence numbers
- Confirm both sides ready

🔹 2 RELIABLE DATA TRANSFER

TCP uses:

- Sequence numbers

- Acknowledgment numbers
- Retransmission on timeout

```
Sender                   Receiver
  |---- pkt1 -----> |
  |<--- ACK1 ------ |
```

If ACK not received → retransmit.

◆ 3 FLOW CONTROL

Uses Receiver Window (RWND)

Formula:

Effective Window = min(CWND, RWND)

Prevents receiver overflow.

◆ 4 CONGESTION CONTROL

Two phases:

Slow Start → Exponential growth
Congestion Avoidance → Linear growth

Graph:

```
CWND
 ^
 |        /---------
 |       /
 |      /
 |     /
 |____/________________> Time
```

◆ 5 CONNECTION TERMINATION

4-Way Handshake:

```
Client                   Server
  |---- FIN -------> |
  |<--- ACK -------- |
```

```
  |<--- FIN -------- |
  |---- ACK -------> |
Closed
```

🔹 STEP-3: IMPORTANT FORMULAS

Transmission Delay = L / R

Throughput ≈ CWND / RTT

Efficiency (Stop-Wait) = (L/R) / (RTT + L/R)

🔹 STEP-4: NUMERICAL

Given:

RTT = 100 ms
MSS = 1000 bytes
CWND = 10 MSS

Throughput ≈ CWND / RTT

= (10 × 1000 bytes) / 0.1 sec
= 10000 / 0.1
= 100,000 bytes/sec

🔹 STEP-5: FINAL ANSWER

👉 TCP is a reliable, connection-oriented transport protocol that ensures ordered, error-free, congestion-controlled data delivery.

🔹 STEP-6: EXAM TIP

- Draw 3 diagrams: handshake, congestion graph, termination.
- Mention 5 TCP features.
- Write formula for throughput.

🔹 Q2. (Chapter-4: Network Layer – Data Plane)

Explain Router Architecture in detail. Discuss input ports, switching fabric, output ports and queueing.

🔹 STEP-1: THEORY

Router = Traffic controller of Internet.

Main Components:

1. Input Ports
2. Switching Fabric
3. Output Ports
4. Routing Processor

◆ STEP-2: FULL DIAGRAM

```
              +----------------------+
              |  Routing Processor   |
              +----------------------+


Input Port 1           Input Port 2
     |                      |
     v                      v
  +------------------------------+
  |        Switching Fabric      |
  +------------------------------+
     ^                      ^
     |                      |
Output Port 1          Output Port 2
```

◆ Explanation

Input Port:

- Receives packet
- Performs longest prefix match

Switching Fabric:

- Transfers packet internally

Output Port:

- Queues packets
- Schedules transmission

Routing Processor:

- Runs routing algorithm

🔹 Switching Types

1. Memory
2. Bus
3. Crossbar

🔹 Queueing

Traffic Intensity (ρ):

ρ = (Arrival Rate × Packet Size) / Bandwidth

If $\rho \to 1 \to$ Delay increases.

🔹 NUMERICAL

Arrival rate = 100 packets/sec
Packet size = 1000 bits
Bandwidth = 100,000 bits/sec

ρ = (100 × 1000) / 100,000
= 1

High congestion.

🔹 FINAL ANSWER

👉 Router architecture includes input processing, internal switching, output scheduling and routing control.

🔹 EXAM TIP

- Draw full router diagram.
- Mention HOL blocking.
- Write switching rate formula.

🔹 Q3. (Chapter-2: Application Layer)

Explain DNS in detail including hierarchy, query types and caching.

🔹 STEP-1: THEORY

DNS converts domain name → IP address.

Hierarchical structure.

◆ STEP-2: DIAGRAM

```
           Root DNS
               |
         -------------
         |           |
       .com        .org
         |
   Authoritative DNS
         |
      www.site.com
```

◆ Query Types

1. Recursive Query
2. Iterative Query

◆ Caching

Local DNS stores resolved addresses.

Reduces delay.

◆ FORMULA

Total DNS Delay = RTT1 + RTT2 + RTT3

◆ NUMERICAL

If:

RTT1 = 40ms
RTT2 = 50ms
RTT3 = 60ms

Total = 150 ms

◆ FINAL ANSWER

👉 DNS is hierarchical distributed system for name resolution.

🔹 EXAM TIP

- Draw hierarchy.
- Mention port 53.
- Explain caching advantage.

🔹 Q4. (Chapter-1: Introduction)

Explain Packet Switching vs Circuit Switching in detail with comparison and numerical.

🔹 DIAGRAM

Circuit:

```
A ====== B ====== C
```

Packet:

```
A → R1 → R2 → C
```

🔹 Comparison

Feature	Circuit	Packet
Resource	Reserved	Shared
Efficiency	Low	High
Delay	Fixed	Variable

🔹 NUMERICAL

Link = 1 Gbps
User need = 100 Mbps

Circuit users = 10

Packet switching:
If active 10% time → More users possible.

🔹 FINAL ANSWER

👉 Packet switching is efficient for bursty data.

🔹 EXAM TIP

- Draw both diagrams.
- Write 4 comparison points.

🔹 Q5. (Chapter-3: Transport Layer)

Explain Congestion Control in detail including Slow Start, Congestion Avoidance and Fast Recovery.

🔹 DIAGRAM

```
CWND
 ^
 |         /---------
 |        /
 |       /
 |______/________________> Time
```

🔹 Phases

1. Slow Start
2. Congestion Avoidance
3. Fast Retransmit
4. Fast Recovery

🔹 FORMULA

Slow Start:

CWND = CWND × 2

Congestion Avoidance:

CWND = CWND + 1 MSS

🔹 NUMERICAL

Initial CWND = 1
RTT1 = 1

RTT2 = 2
RTT3 = 4
RTT4 = 8

If loss at 8:

SSTHRESH = 4
CWND reset.

◆ FINAL ANSWER

👉 TCP congestion control prevents network overload using dynamic window adjustment.

◆ EXAM TIP

- Draw CWND vs Time graph.
- Mention exponential & linear growth.
- Write SSTHRESH rule.

◆ Q6. (Chapter-3: Transport Layer)

Explain Reliable Data Transfer Protocol (rdt1.0, rdt2.0, rdt2.1, rdt3.0) in detail with diagrams and limitations.

◆ STEP-1: THEORY (FROM ZERO LEVEL)

✅ Intuition

When sending data over network:

- Bits may flip (errors)
- Packets may be lost
- ACKs may be corrupted

So we improve protocol step-by-step.

🔹 1️⃣ rdt1.0 (Perfect Channel)

Assumption:

No loss, no errors.

```
Sender -------- pkt --------> Receiver
```

Problem:
Unrealistic in real networks.

🔹 2️⃣ rdt2.0 (Bit Errors Present)

Uses:

- Checksum
- ACK
- NAK

```
Sender ---- pkt ----> Receiver
       <---- ACK ----
```

Problem:
ACK/NAK may be corrupted.

🔹 3️⃣ rdt2.1 (Fixes ACK corruption)

Adds:

- Sequence numbers (0,1)

```
Sender (Seq=0) ----> Receiver
         <---- ACK0
```

Now duplicates detected.

◆ 4️⃣ rdt3.0 (Handles Packet Loss)

Adds:

- Timer
- Retransmission

```
Sender ---- pkt0 ----> Receiver
(No ACK received)
Timeout → Retransmit
```

◆ STEP-2: SUMMARY DIAGRAM

```
rdt1.0 → Perfect Channel
rdt2.0 → Errors
rdt2.1 → Errors + Corrupted ACK
rdt3.0 → Errors + Loss + Timeout
```

◆ STEP-3: FORMULA

Stop-and-Wait Efficiency:

U = (L/R) / (RTT + L/R)

◆ STEP-4: NUMERICAL

Given:

L = 1000 bits
R = 1 Mbps
RTT = 100 ms

L/R = 0.001 sec

U = 0.001 / 0.101
≈ 0.0099 (0.99%)

Very inefficient.

◆ STEP-5: FINAL ANSWER

👉 rdt evolved step-by-step to handle errors, corruption and packet loss using sequence numbers and timers.

◆ STEP-6: EXAM TIP

- Draw evolution flow.
- Mention fatal flaw in rdt2.0.
- Write stop-and-wait formula.

◆ Q7. (Chapter-4: Network Layer – Data Plane)

Explain Longest Prefix Matching in detail with example and calculation.

◆ STEP-1: THEORY

Router forwards packet based on destination IP.

If multiple matches → choose most specific prefix.

◆ STEP-2: FORWARDING TABLE

```
Prefix              Interface
1100                0
110010              1
11001010            2
Otherwise           3
```

Destination IP:

110010101100

Matching:

1100 → Yes
110010 → Yes
11001010 → Yes

Longest = 11001010

Interface = 2

◆ STEP-3: FORMULA

Match = Number of identical leading bits.

🔹 STEP-4: NUMERICAL

Destination:

110010110000

Matches:

1100 → Yes
110010 → Yes
11001010 → No

Longest = 110010

Interface = 1

🔹 STEP-5: FINAL ANSWER

👉 Router selects forwarding entry with maximum matching prefix bits.

🔹 STEP-6: EXAM TIP

- Show bit comparison.
- Highlight longest match.
- Draw table clearly.

🔹 Q8. (Chapter-2: Application Layer)

Explain HTTP in detail including request/response format, cookies and response time.

🔹 STEP-1: THEORY

HTTP = Web protocol.

Uses TCP.

Stateless protocol.

🔹 HTTP REQUEST FORMAT

```
GET /index.html HTTP/1.1
Host: www.site.com
```

```
User-Agent: Chrome
```

◆ HTTP RESPONSE FORMAT

```
HTTP/1.1 200 OK
Content-Type: text/html
Content-Length: 1000
```

◆ COOKIES

Used to maintain state.

```
Client → Server
Server sends Cookie ID
Client stores
Client sends back next time
```

◆ RESPONSE TIME

Non-persistent:

Response Time = 2RTT + Transmission

Persistent:

≈ 1RTT + Transmission

◆ NUMERICAL

RTT = 100 ms
Transmission = 50 ms

Non-persistent:

= 2×100 + 50
= 250 ms

Persistent:

= 100 + 50
= 150 ms

🔹 FINAL ANSWER

👉 HTTP is stateless application-layer protocol using request-response model over TCP.

🔹 EXAM TIP

- Draw request-response diagram.
- Write 200 OK.
- Mention cookies.

🔹 Q9. (Chapter-4: Network Layer – Data Plane)

Explain Switching Fabric types in detail with comparison.

🔹 STEP-1: THEORY

Switching fabric transfers packets inside router.

🔹 TYPES

1️⃣ Memory-Based

```
Input → CPU Memory → Output
```

Limited by memory speed.

2️⃣ Bus-Based

```
Inputs → Shared Bus → Outputs
```

Bus contention possible.

3️⃣ Crossbar

```
Input1  X  Output1
Input2  X  Output2
```

Parallel switching.

🔹 FORMULA

Switching Capacity $\geq N \times$ Line Rate

🔹 NUMERICAL

Ports = 8
Line rate = 1 Gbps

Switching capacity required = 8 Gbps

🔹 FINAL ANSWER

👉 Switching fabric moves packets using memory, bus or interconnection networks.

🔹 EXAM TIP

- Draw all 3 types.
- Mention scalability.
- Write switching rate formula.

🔹 Q10. (Chapter-3: Transport Layer)

Explain Flow Control and Congestion Control difference in detail with diagram and formula.

🔹 STEP-1: THEORY

Flow Control:

Protects receiver.

Congestion Control:

Protects network.

🔹 DIAGRAM

Flow Control:

```
Sender → Receiver (Limited by RWND)
```

Congestion Control:

```
Sender → Network → Receiver (Limited by CWND)
```

🔹 FORMULA

Effective Window = min(CWND, RWND)

🔹 NUMERICAL

CWND = 10
RWND = 6

Effective = 6

Sender sends only 6 segments.

🔹 FINAL ANSWER

👉 Flow control prevents receiver overflow; congestion control prevents network overload.

🔹 EXAM TIP

- Draw two small diagrams.
- Write difference clearly.
- Mention min(CWND, RWND).

📘 CHAPTER 1 — INTRODUCTION

🟢 1️⃣ TCP/IP Layers Memory Trick

Layers:

Application
Transport
Network
Link

🧠 **Trick:**

👉 **"All Teachers Need Lunch"**

A → Application
T → Transport
N → Network
L → Link

If exam asks layers → Just remember “Teachers Need Lunch” 😄

🟢 2️⃣ Four Types of Delay

Processing
Queueing
Transmission
Propagation

🧠 **Trick:**

👉 **"Please Quit Talking Politics"**

P → Processing
Q → Queueing
T → Transmission
P → Propagation

🟢 3️⃣ Packet vs Circuit Switching

Circuit = Reserved
Packet = Shared

🧠 **Trick:**

👉 **C = Call (Telephone)**
👉 **P = Packets (Internet)**

If it looks like phone → Circuit
If it looks like Internet → Packet

🟢 4️⃣ Bottleneck Link

Throughput = Minimum link capacity

🧠 **Trick:**

👉 **"Smallest Pipe Controls Water"**

Always choose minimum.

🌐 CHAPTER 2 — APPLICATION LAYER

🟢 5️⃣ Client-Server vs P2P

Client-Server = Centralized
P2P = Distributed

🧠 **Trick:**

👉 **C = Central**
👉 **P2P = People to People**

🟢 6️⃣ DNS Hierarchy

Root → TLD → Authoritative

🧠 **Trick:**

👉 **"Rich Teachers Are Awesome"**

R → Root
T → TLD
A → Authoritative

🟢 7️⃣ HTTP Request Methods

GET
POST
HEAD
PUT

🧠 **Trick:**

👉 **"Good People Have Power"**

G → GET
P → POST
H → HEAD
P → PUT

🟢 8️⃣ Transport Requirements of Apps

Integrity
Timing
Throughput
Security

🧠 Trick:

👉 **"I Trust The System"**

I → Integrity
T → Timing
T → Throughput
S → Security

🟢 9️⃣ HTTP Response Time

Non-persistent:

2RTT + Transmission

🧠 Trick:

👉 **"Open → Ask → Reply → Close" = 2 Trips**

🔄 CHAPTER 3 — TRANSPORT LAYER

🟢 🔟 TCP 3-Way Handshake

SYN
SYN-ACK
ACK

🧠 Trick:

👉 **"S-S-A" = Say Sorry Again"**

SYN → Start
SYN-ACK → Response
ACK → Confirm

🟢 1️⃣ 1️⃣ TCP 4-Way Termination

FIN
ACK
FIN
ACK

🧠 Trick:

👉 **"F-A-F-A" = Finish And Finish Again**

🟢 1️⃣ 2️⃣ TCP Congestion Phases

Slow Start
Congestion Avoidance
Fast Recovery

🧠 Trick:

👉 **"Start → Avoid → Fix"**

🟢 1️⃣ 3️⃣ CWND Growth

Slow Start → Double
Avoidance → +1

🧠 Trick:

👉 **"Double First, Then Gentle"**

🟢 1️⃣ 4️⃣ Reliable Data Transfer Evolution

rdt1.0 → Perfect
rdt2.0 → Errors
rdt2.1 → Errors + Sequence
rdt3.0 → Loss + Timer

🧠 Trick:

👉 **"Perfect → Problem → Sequence → Timer"**

🟢 1️⃣ 5️⃣ Stop-and-Wait Efficiency Formula

U = (L/R) / (RTT + L/R)

🧠 Trick:

👉 **"Send Time / Total Time"**

🟢 1️⃣ 6️⃣ Flow vs Congestion Control

Flow → Receiver
Congestion → Network

🧠 Trick:

👉 **"F → Friend (Receiver)"**
👉 **"C → Crowd (Network)"**

🟢 1️⃣ 7️⃣ UDP Features

Connectionless
Unreliable
No congestion control
Fast

🧠 Trick:

👉 **"CUFF"**

C → Connectionless
U → Unreliable
F → Fast
F → Few features

🟢 1️⃣ 8️⃣ TCP vs UDP

TCP = Reliable
UDP = Fast

🧠 Trick:

👉 **T = Trustworthy**
👉 **U = Ultra Fast**

🌍 CHAPTER 4 — NETWORK LAYER

🟢 1️⃣ 9️⃣ Router Components

Input
Switching Fabric
Output
Routing Processor

🧠 Trick:

👉 **"I See Our Router"**

I → Input
S → Switching
O → Output
R → Routing

🟢 2️⃣ 0️⃣ Switching Fabric Types

Memory
Bus
Crossbar

🧠 Trick:

👉 **"My Bus Crossed"**

M → Memory
B → Bus
C → Crossbar

🟢 2️⃣ 1️⃣ Output Scheduling

FIFO
Priority
Round Robin

🧠 Trick:

👉 **"First People Rotate"**

F → FIFO
P → Priority
R → Round Robin

🟢 2️⃣ 2️⃣ IP Datagram Important Fields

Version
IHL
TTL
Protocol
Source
Destination

🧠 **Trick:**

👉 **"Very Intelligent Teachers Prefer Smart Decisions"**

V → Version
I → IHL
T → TTL
P → Protocol
S → Source
D → Destination

🟢 2️⃣ 3️⃣ Longest Prefix Matching

Most specific prefix wins.

🧠 **Trick:**

👉 **"Longest Wins"**

Always choose maximum matching bits.

🟢 2️⃣ 4️⃣ NAT

Private → Public

🧠 **Trick:**

👉 **"NAT = New Address Translation"**

🟢 2️⃣ 5️⃣ Traffic Intensity

ρ = (Arrival Rate × Packet Size) / Bandwidth

🧠 **Trick:**

👉 **"Load / Capacity"**

If ρ near 1 → Danger ⚠️

🔥 SUPER QUICK REVISION TABLE

Topic	Memory Key
TCP Handshake	SSA
Termination	FAFA
Delays	PQTP
DNS	RTA
Switching	MBC
Scheduling	FPR
Layers	ATNL
Congestion	Double then Linear
Throughput	Smallest Pipe
Flow vs Congestion	Friend vs Crowd

📘 CHAPTER 1 — INTRODUCTION (Must Know 100%)

🔥 1️⃣ TCP/IP Layers (Very Important)

👉 4 Layers:

Application
Transport
Network
Link

🧠 Remember: **ATNL**

🔥 2️⃣ Four Types of Delay

Formula:

D = dproc + dqueue + dtrans + dprop

Where:

Transmission = L / R
Propagation = Distance / Speed

⚠️ Examiner LOVES this numerical.

🔥 3️⃣ Throughput & Bottleneck

Throughput = Minimum link capacity

🧠 Always choose smallest link.

🔥 4️⃣ Packet vs Circuit Switching

Packet	Circuit
Shared	Reserved
Internet	Telephone

⚠️ Common 5 or 10 mark question.

🌐 CHAPTER 2 — APPLICATION LAYER

🔥 5️⃣ Client-Server vs P2P

Client-Server:

Centralized
Server always ON

P2P:

No central server
Peers communicate directly

Draw diagram in exam.

🔥 6️⃣ HTTP (Very Very Important)

Non-persistent:

Response Time = 2RTT + Transmission

Persistent:

≈ 1RTT + Transmission

HTTP is **Stateless**

Cookies maintain state.

🔥 7️⃣ DNS Hierarchy

Root
TLD
Authoritative

Port: 53

Total Delay = RTT1 + RTT2 + RTT3

Draw tree diagram always.

🔥 8️⃣ Web Caching

Reduces:

Delay
Bandwidth usage

Hit Ratio = Cache Hits / Total Requests

🔄 CHAPTER 3 — TRANSPORT LAYER (MOST IMPORTANT CHAPTER)

⭐ ⭐ ⭐ MOST IMPORTANT CHAPTER ⭐ ⭐ ⭐

🔥 9️⃣ TCP 3-Way Handshake (Always asked)

SYN
SYN-ACK
ACK

Draw diagram always.

🔥 🔟 TCP Termination

FIN
ACK
FIN
ACK

Full duplex closing.

🔥 1️⃣ 1️⃣ Reliable Data Transfer Evolution

rdt1.0 → Perfect
rdt2.0 → Errors
rdt2.1 → Seq numbers
rdt3.0 → Timer

Stop-and-Wait Efficiency:

U = (L/R) / (RTT + L/R)

Examiner LOVES this.

🔥 1️⃣ 2️⃣ Sliding Window

Window Size = Bandwidth × RTT

Improves efficiency.

🔥 1️⃣3️⃣ Flow Control

Effective Window = min(CWND, RWND)

Flow → Protect receiver
Congestion → Protect network

🔥 1️⃣4️⃣ TCP Congestion Control

Slow Start → Exponential
Congestion Avoidance → Linear

SSTHRESH = CWND / 2 after loss

Draw CWND vs Time graph.

Very important for 10 marks.

🔥 1️⃣5️⃣ UDP

Connectionless
Unreliable
No congestion control
Header size = 8 bytes

Draw header format if asked.

🔥 1️⃣6️⃣ Internet Checksum

1's complement addition
16-bit words
Wrap-around carry

Binary example often asked.

🌍 CHAPTER 4 — NETWORK LAYER (DATA PLANE)

🔥 1️⃣7️⃣ Router Architecture

Input Port
Switching Fabric

Output Port
Routing Processor

Draw full block diagram.

🔥 1️⃣ 8️⃣ Switching Fabric Types

Memory
Bus
Crossbar

Switching Capacity ≥ N × Line Rate

🔥 1️⃣ 9️⃣ Longest Prefix Matching

Choose most specific match.

Draw forwarding table example.

🔥 2️⃣ 0️⃣ Output Queuing

Traffic Intensity:

ρ = (Arrival Rate × Packet Size) / Bandwidth

If $\rho \rightarrow 1$
Delay increases rapidly.

🔥 2️⃣ 1️⃣ NAT

Private IP → Public IP
Uses Port Mapping

Draw NAT table.

🔥 2️⃣ 2️⃣ IP Datagram Format

Important Fields:

Version
IHL
TTL
Protocol

Source IP
Destination IP

TTL prevents infinite looping.

Draw header format.

🚨 MOST REPEATED EXAM NUMERICALS

✓ Transmission Delay = L / R
✓ Propagation Delay = Distance / Speed
✓ Stop-and-Wait Efficiency
✓ Window Size = B × RTT
✓ Throughput = Min link
✓ Traffic Intensity (ρ)
✓ HTTP Response Time

If you remember these → 30% paper solved.

🎯 LAST NIGHT STRATEGY (VERY IMPORTANT)

🟢 Step 1

Revise:

TCP handshake
Congestion control graph
Router architecture
DNS hierarchy
IP header

These are guaranteed marks.

🟢 Step 2

Practice:

1 Stop-and-Wait numerical
1 Delay numerical
1 Throughput problem

🟢 Step 3

Memorize:

All formulas
All diagrams

⚠️ COMMON MISTAKES TO AVOID

❌ Mixing propagation & transmission delay
❌ Forgetting min(CWND, RWND)
❌ Drawing OSI model instead of TCP/IP
❌ Not drawing diagram in 5-mark question
❌ Not writing formula before calculation

🎯 SAFE SCORE STRATEGY

If paper is difficult:

Attempt:

1. TCP
2. Congestion Control
3. Router Architecture
4. DNS
5. Delay Numerical

These topics almost always come.

🧠 60-SECOND QUICK RECALL BEFORE EXAM

Say this mentally:

ATNL
PQTP
SSA
FAFA
Double then Linear
Min(CWND,RWND)
Longest Wins

Smallest Pipe
ρ formula

If you can recall these → You are ready.

💪 FINAL MOTIVATION

You now have:

✓ 30 Important 5-Mark Questions
✓ 20 Important 10-Mark Questions
✓ Memory Tricks
✓ Full Numerical Practice
✓ Ultra Revision Sheet

You are **well prepared**.

🚨 ONE PAGE EMERGENCY CHEAT SHEET

(Read 10 Minutes Before ACN Exam)

⚠️ Only from your 4 Chapters
⚠️ Only formulas + diagrams + keywords
⚠️ Maximum marks, minimum reading

📘 CHAPTER 1 — INTRODUCTION

✅ TCP/IP Layers (Write in Order)

```
Application
Transport
Network
Link
```

🧠 Trick: **ATNL**

✅ 4 Types of Delay

Formula:

$$
D = d_{proc} + d_{queue} + d_{trans} + d_{prop}
$$

Where:

- $d_{trans} = L / R$
- $d_{prop} = Distance / Speed$

⚠️ Transmission ≠ Propagation

✅ Throughput

$$
Throughput = \min (Link\ Capacities)
$$

🧠 Smallest Pipe Rule

✅ Packet vs Circuit

Packet → Shared
Circuit → Reserved

🌐 CHAPTER 2 — APPLICATION LAYER

✅ Client-Server vs P2P Diagram

Client-Server:

```
Clients → Server ← Clients
```

P2P:

```
Peer ↔ Peer ↔ Peer
```

✅ HTTP Response Time

Non-persistent:

[
2RTT + Transmission
]

Persistent:

[
1RTT + Transmission
]

HTTP = Stateless
Cookies = Maintain State

✅ DNS Hierarchy

```
Root
  ↓
TLD (.com)
  ↓
Authoritative
```

Port: **53**

✅ Cache Hit Ratio

[
Hit\ Ratio = \frac{Cache\ Hits}{Total\ Requests}
]

🔄 CHAPTER 3 — TRANSPORT (VERY IMPORTANT)

✅ TCP 3-Way Handshake

```
SYN
SYN-ACK
ACK
```

✅ TCP Termination

```
FIN
ACK
FIN
ACK
```

✅ Stop-and-Wait Efficiency

$$
U = \frac{L/R}{RTT + L/R}
$$

✅ Sliding Window

$$
Window\ Size = Bandwidth \times RTT
$$

✅ Flow vs Congestion

$$
Effective\ Window = \min(CWND, RWND)
$$

Flow → Receiver
Congestion → Network

✅ TCP Congestion Control

Slow Start → Double CWND
Congestion Avoidance → +1 MSS

After loss:

$$
SSTHRESH = CWND / 2
$$

Draw CWND vs Time graph if asked.

✅ UDP

- Connectionless
- Unreliable
- Header = 8 bytes

✅ Internet Checksum

- 16-bit words
- 1's complement addition
- Wrap-around carry

🌍 CHAPTER 4 — NETWORK LAYER

✅ Router Architecture

```
Input → Switching Fabric → Output
        ↑
   Routing Processor
```

✅ Switching Fabric Types

Memory
Bus
Crossbar

[
Switching\ Rate \ge N \times Line\ Rate
]

✅ Longest Prefix Matching

Choose most matching leading bits.

🧠 Longest Wins.

✅ Traffic Intensity

$$
\rho = \frac{Arrival\ Rate \times Packet\ Size}{Bandwidth}
$$

If $\rho \to 1$
Delay → Very high

✅ NAT

Private IP → Public IP
Uses Port Mapping

✅ IP Datagram Important Fields

Version
IHL
TTL
Protocol
Source
Destination

TTL prevents infinite looping.

🚨 MOST LIKELY NUMERICALS

✓ Transmission Delay
✓ Propagation Delay
✓ Stop-and-Wait Efficiency
✓ Window Size (B × RTT)
✓ Throughput (Min link)
✓ Traffic Intensity (ρ)
✓ HTTP Response Time

🧠 30-SECOND RECALL FORMULA LIST

```
D = dproc + dqueue + dtrans + dprop
dtrans = L/R
dprop = Distance/Speed
Throughput = Min link
U = (L/R)/(RTT + L/R)
Window = B × RTT
```

```
Effective = min(CWND, RWND)
ρ = (λL)/R
SSTHRESH = CWND/2
```

🎯 IF YOU FORGET EVERYTHING

Write about:

TCP
Congestion Control
Router Architecture
DNS
Delay Formula

These 5 topics cover most paper.